

The statistical learning framework can be formulated as

$$(\mathcal{X}, \mathcal{Y}, \mathcal{P}, \mathcal{H}, \mathcal{L})$$

where

- $\mathcal{X}$ is the input space,
- $\mathcal{Y}$ is the output space,
- $\mathcal{P}$ is the probability distribution over $\mathcal{X} \times \mathcal{Y}$,
- $\mathcal{H}$ is the hypothesis space, and
- $\mathcal{L}$ is the loss function.

The goal is to find a hypothesis $h : \mathcal{X} \rightarrow \mathcal{Y}$, where $h \in \mathcal{H}$, that minimizes the expected loss

$$\mathbb{E}_{(x,y) \sim \mathcal{P}} [\mathcal{L}(h(x), y)]$$

There is a training set $S = \{(x_i, y_i) : x_i \in \mathcal{X}, y_i \in \mathcal{Y}\}_{i=1}^n$ drawn from the distribution $\mathcal{P}$. The [Empirical Risk Minimization] (ERM) principle is often used, which minimizes the empirical loss over the training set S .

0.0.1 See also

0.0.2 References